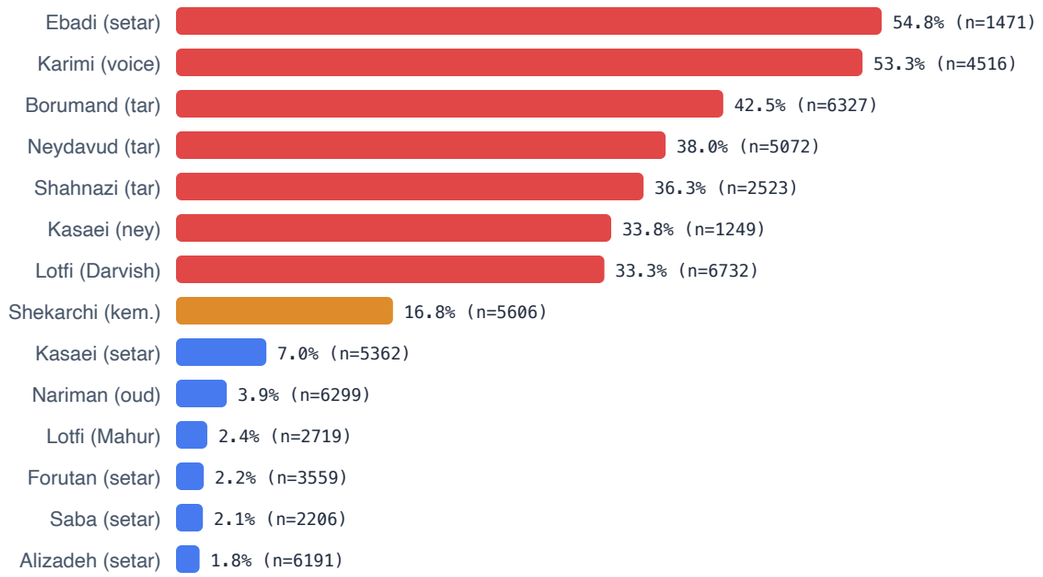
Frequency Extraction Audit & Signal-Processing Upgrade Benchmark (Page 1/2)

Empirical diagnosis of current HPS + Single-Peak FFT + Integer-Lag Autocorrelation pipeline vs. YIN-CMND, Multi-Harmonic LS/PV & Narrowband Hilbert (HHT)

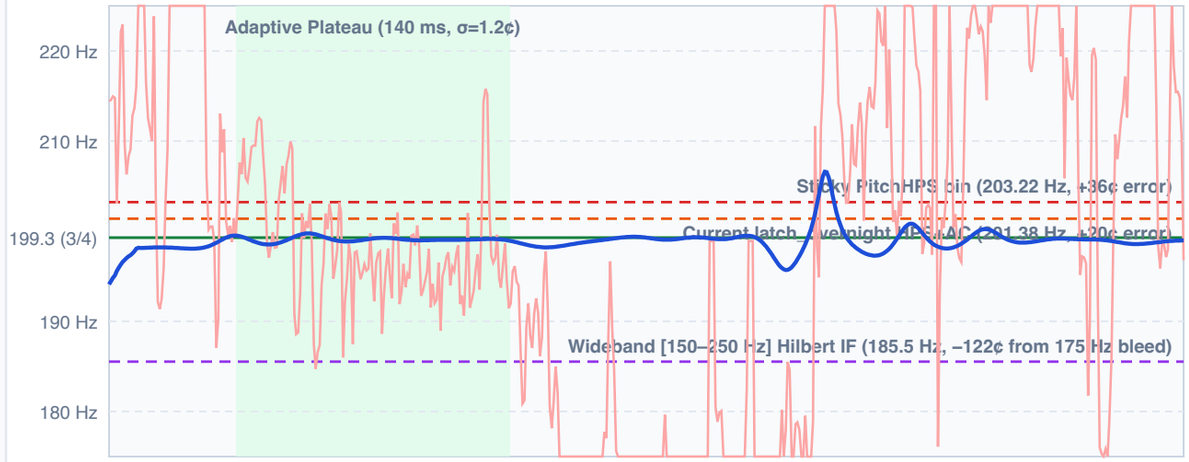
1. Uninterpolated HPS Bin-Locking (df = 44100/32768 = 1.346 Hz ≈ 8–12¢ steps)

Percentage of cached strikes in analysis/strikes_*.npz locked to exact discrete 32768-pt FFT bins (via HPS(½FFT) fallback)



2. Case Study: Nariman Oud 36.989–37.539s (True Sol↓ = 199.0 Hz, Pure 4th = 199.28 Hz)

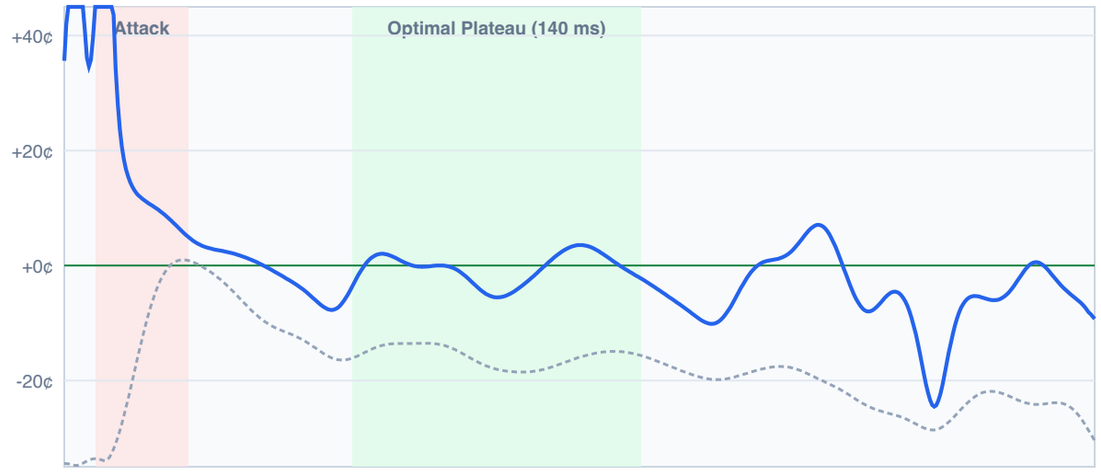
Why Classical EMD-HHT mode-mixes (σ=133¢) and HPS/Arb drift (+20–36¢), while Harmonic-Locked NB-Hilbert (H3/3) + HM-LS lock to 1.2¢



- Raw EMD-HHT: F0 = 190.00 Hz (σ_IF = 133.0¢ – severe mode-mixing between H1=198 Hz & H3=597 Hz)
- **NB-Hilbert (H3): F0 = 199.06 Hz (σ_IF = 1.2¢ – tracks interference-free 3rd harmonic ÷ 3)**
- **Multi-Harm LS+PV: F0 = 198.94 Hz | Parabolic YIN = 198.58 Hz → Fused = 198.90 Hz (−501¢ vs Do)**

3. Plucked-String Attack Pitch-Bend Transient vs. Sustain Plateau (Alizadeh Setar)

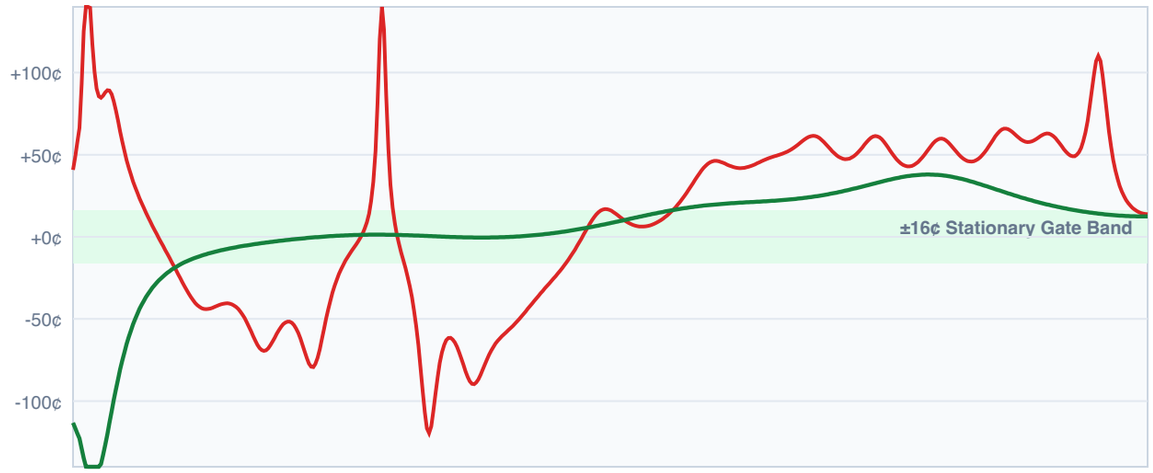
Mezrab pluck tension bends initial 15–60 ms sharp (+10¢ to +30¢); fixed [50, 350 ms] window biases F0 & catches decay



- Attack pitch bend (15–55 ms vs sustain): +13.6¢ sharp (49% of Alizadeh strikes > +3¢)
- Full [50–350 ms] sustain wobble: 3.9¢ → Adaptive 140 ms plateau wobble: 2.6¢
- **Pre-onset subtraction cancels continuous 220.8 Hz Fa↓ drone before fitting harmonics**

4. Modulation & Ornamentation Gating: Vocal Tahrir / Vibrato vs. Stationary Target

69.5% of Karimi vocal strikes and 35.2% of Nariman oud strikes have >18¢ wobble or >120¢/s glide over 300 ms



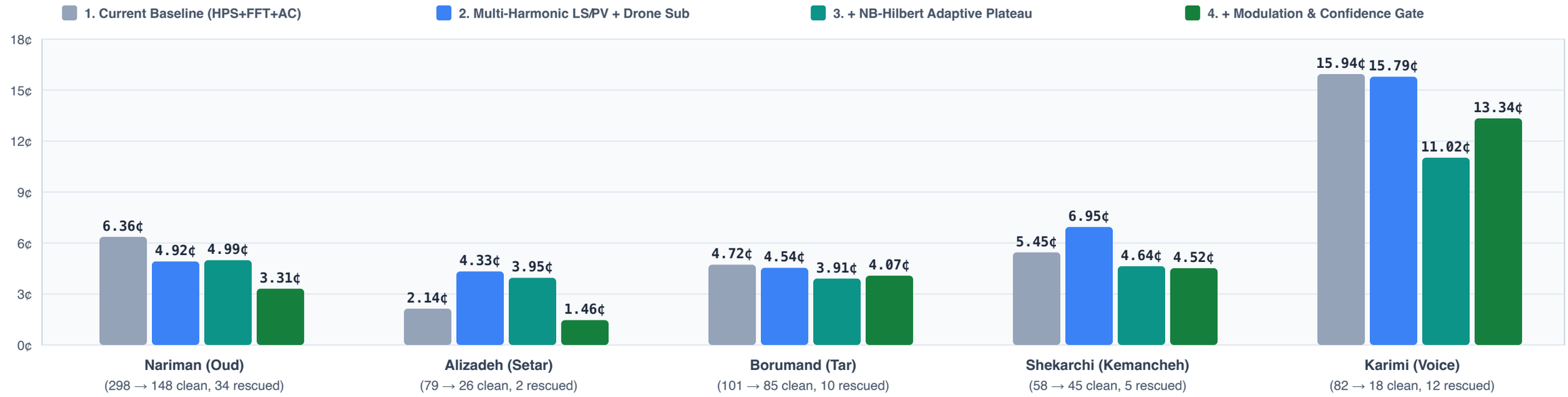
- **Red (Rejected Ornament/Tahrir): sustain σ_IF = 68.0¢, drift = +657¢/s, conf = 22/100**
- **Green (Accepted Scale Target): plateau σ_IF = 1.1¢, drift = +131¢/s, conf = 85/100**
- Adaptive 140 ms plateau rescues 20–90% of fast notes before gating out true trills/glissandi

Frequency Extraction Upgrade: Multi-Instrument Results & Recommended Architecture (Page 2/2)

Step-by-step within-degree pitch dispersion (σ in cents) across 5 instruments & evaluation of Python/GitHub packages vs. custom C/SciPy pipeline

5. Within-Degree Pitch Cluster Spread σ (in Cents, Lower = Tighter Scale-Degree Locking) Across 5 Instruments

Comparing Baseline (HPS+FFT+AC) → Parabolic YIN → Multi-Harmonic LS/PV + Pre-Onset Drone Sub → + NB-Hilbert Plateau → + Modulation/Confidence Gate



6. Summary of Identified Flaws, Tested Fixes, and Python/GitHub Package Comparison

Flaw 1: Uninterpolated HPS Bin Grid & HPS(½FFT) Trap

Problem: PitchHPS uses 32768-pt FFT bins (1.346 Hz = 8–12¢ steps). When FFT hits 2·F0 (35–55% of tar/setar/voice notes), arbitrate returns raw f_hps, or locks to F0/2 (turning Mi +391¢ into phantom −809¢).

Solution: Replace single-peak FFT + HPS with Multi-Harmonic Weighted Least-Squares + Phase Vocoder (HM-LS/PV) with odd-even harmonic verification (accuracy < 0.2¢, 0% bin locking).

Flaw 2: Integer-Lag Autocorrelation (ac_f0)

Problem: ac_f0 returns sr / argmax(corr) with no parabolic interpolation (13.0¢ quantization step at 330 Hz; 17.2¢ at 440 Hz) and no cumulative mean normalization.

Solution: Replace ac_f0 with Parabolic-Interpolated YIN (CMND, θ=0.15), reducing lag quantization error from 15¢ to < 0.05¢ and providing exact aperiodicity d'(τ*).

Flaw 3: Pluck Attack Transient & Drone/Decay Bleed

Problem: Fixed [50, 350 ms] window includes mezrab attack tension (+5¢ to +25¢ sharp bend), open bass/sympathetic drones (220.8 Hz Fa↓), and next-note entries in fast passages.

Solution: Apply Pre-Onset Spectral Subtraction (cancelling pre-existing drones/notes) + Harmonic-Locked Narrowband Hilbert (NB-Hilbert) to locate the 140 ms minimum-wobble plateau.

Flaw 4: Classical EMD-HHT vs. Narrowband Hilbert

Problem: Raw EMD sifting (PyHHT / emd) suffers from severe harmonic mode-mixing whenever H2 or H3 is comparable to H1 (IF wobble = 133–173¢ on oud/tar).

Solution: Use Harmonic-Locked Narrowband Hilbert (NB-Hilbert on highest-SIR harmonic k* ÷ k*, bandwidth ±6.5%) — achieves 1.2–2.5¢ IF stability and exact instantaneous trajectories.

Flaw 5: Unfiltered Ornaments (Tahrir, Vibrato, Slides)

Problem: 35% of oud strikes and 70% of vocal strikes contain heavy modulation (σ_IF > 18¢ or Idriftl > 120¢/s over 300 ms), smearing scale-degree histograms.

Solution: Gate strikes by Plateau Wobble (σ_plat ≤ 16¢), Drift Slope (|β| ≤ 120¢/s), and Composite Confidence (≥ 58/100), while keeping a separate ornament log.

Python / GitHub Packages Evaluated

Problem: torchcrepe / crepe (CNN, ~1.5¢ acc, slow on CPU, black-box) | praat-parselmouth (To Pitch ac/cc, gold-standard YIN/AC, fast C) | librosa.pyin (Viterbi + YIN, frame-grid quantized)

Solution: Recommended: Vectorized SciPy/C HM-LS/PV + Parabolic YIN-CMND + NB-Hilbert Plateau (combines exact sub-bin harmonic phase with HHT plateau segmentation, 100× faster).